

The Effect of Ascorbyl Palmitate on the PI3K-AKT Pathway Activity in Burkitt's Lymphoma Cells (Raji Cells)

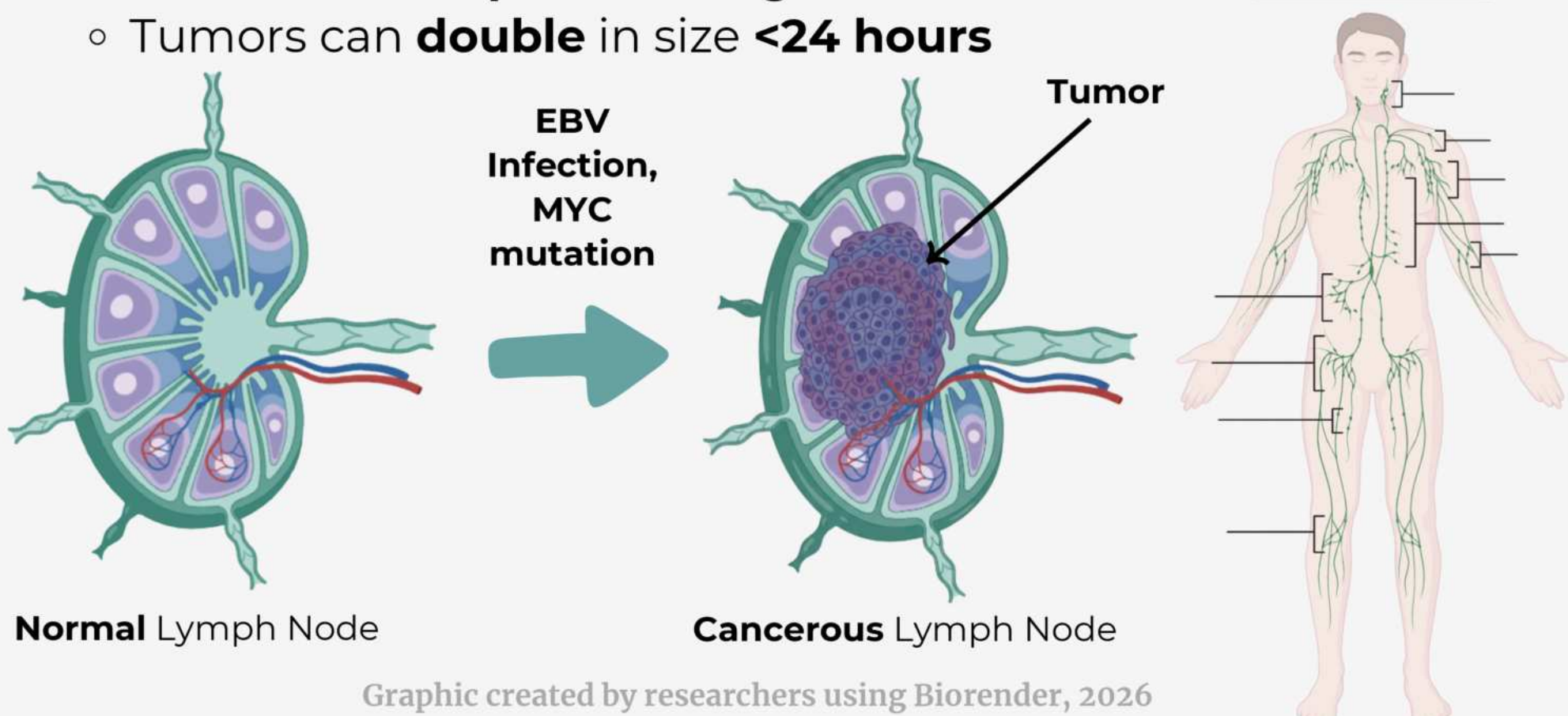
Background

Burkitts Lymphoma

A highly **agressive** non Hodgkins B-cell lymphoma with **rapid tumor growth**.

- Often associated with Epstein-Barr Virus (**EBV**) and Human Immunodeficiency Virus (**HIV**)
- Most prevalent in **children and adolescents** in equatorial Africa and South America
- One of the **fastest proliferating** human cancers
 - Tumors can **double** in size **<24 hours**

Common Lymph Node locations

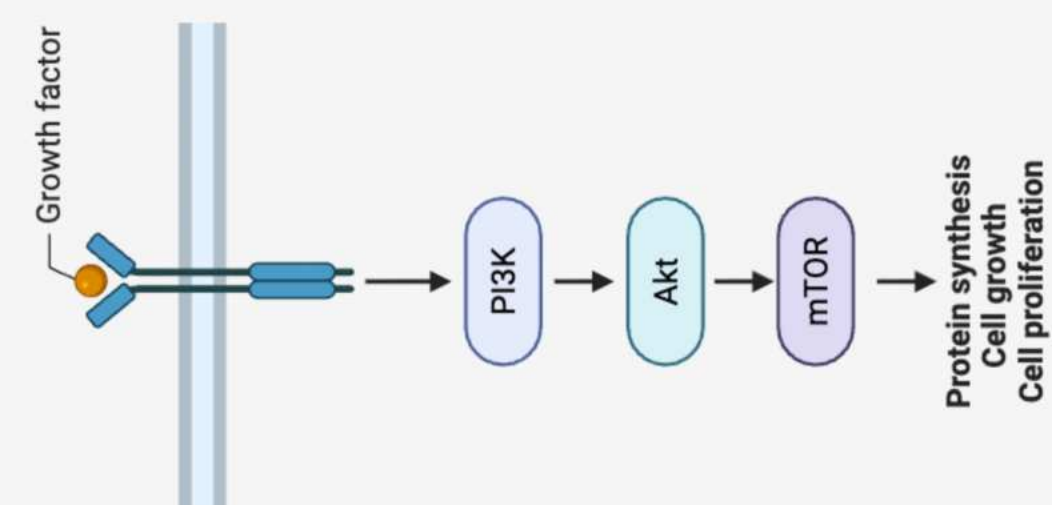


PI3K-AKT Cellular Signaling Pathway

PI3K-AKT Pathway is **Overexpressed** leading to **excessive & unregulated cell growth**

Therefore:

AKT Protein ↑



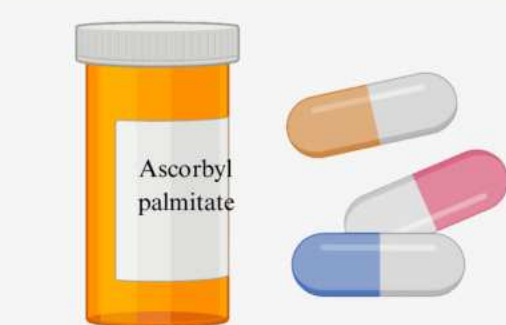
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Current Treatment: Intensive Chemotherapy

- Individuals are most commonly treated with **high dosages** of chemotherapy which is **highly toxic**
- Side effects** include: nausea, hair loss, heart problems, etc.
- If not caught in the early stages then chemotherapy is **ineffective** and can lead to recurrent disease with a **life expectancy <2.8 months**.

Proposed Treatment: Ascorbyl Palmitate

- Fat soluble** derivative of vitamin C
 - easier movement across cell membrane
- Widely approved** as a food antioxidant and used in various skincare products due to its stability
 - safe for human consumption**



Graphic created by researchers using Biorender, 2026

Hypothesis

If:

Raji Cells are treated with varying concentrations of ascorbyl palmitate

Then:

Cell viability & the expression of p-akt will both decrease as concentration increases

Justification:

Because the PI3K-AKT pathway is overexpressed and Ascorbyl palmitate's lipophilic structure allows it to diffuse across the membrane easier



Significance

- Ascorbyl Palmitate is **widely available** as a food antioxidant and already **approved** for human consumption
- Can provide a **safer non-toxic** alternative treatment for Burkitt's Lymphoma
- Novel** approach since Ascorbyl Palmitate has not been tested on Raji cells
- This research can inform future research into **lipophilic derivatives** as cancer therapeutics

Methodology

1. Cell Culture

Raji Cells - Burkitts Lymphoma

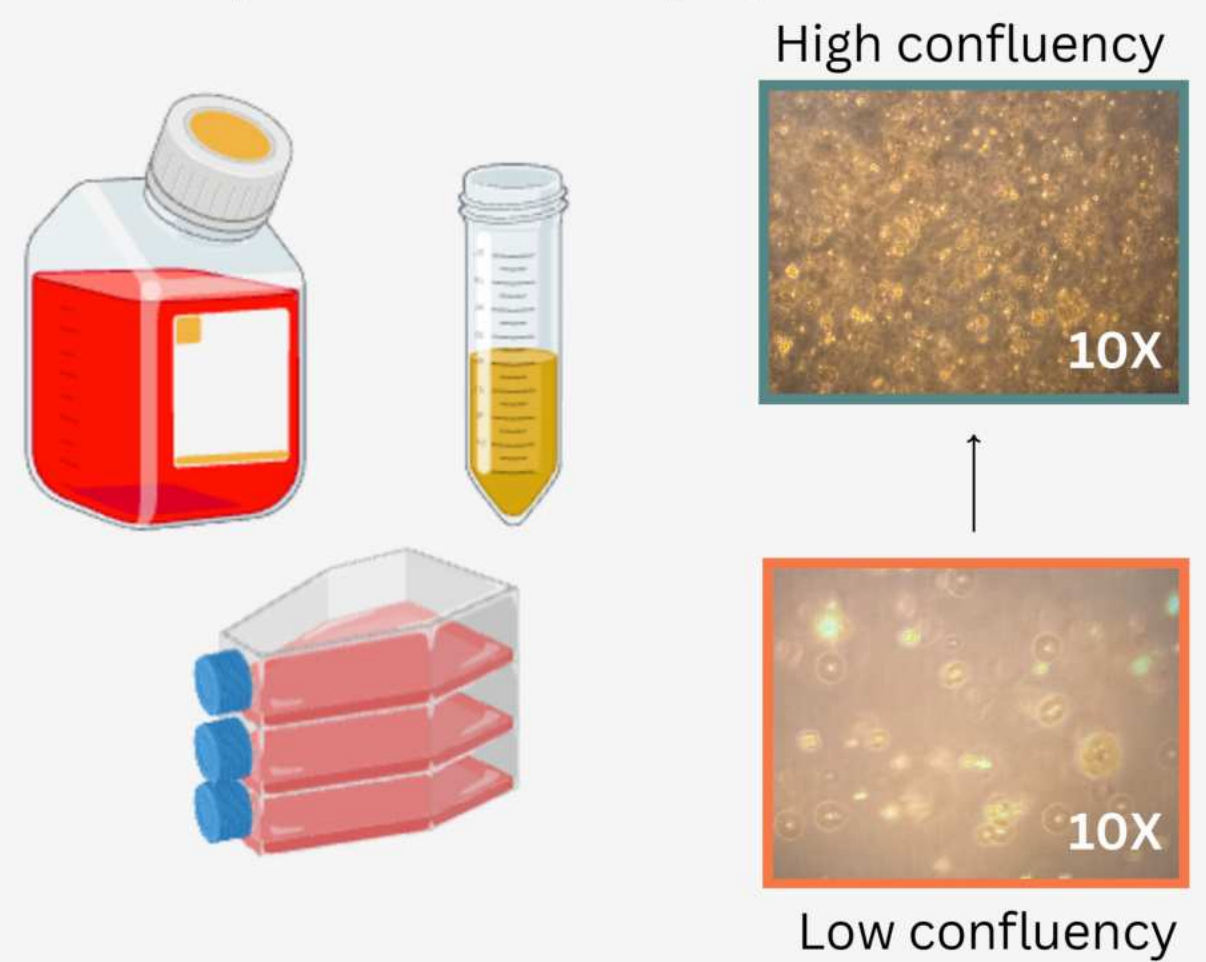
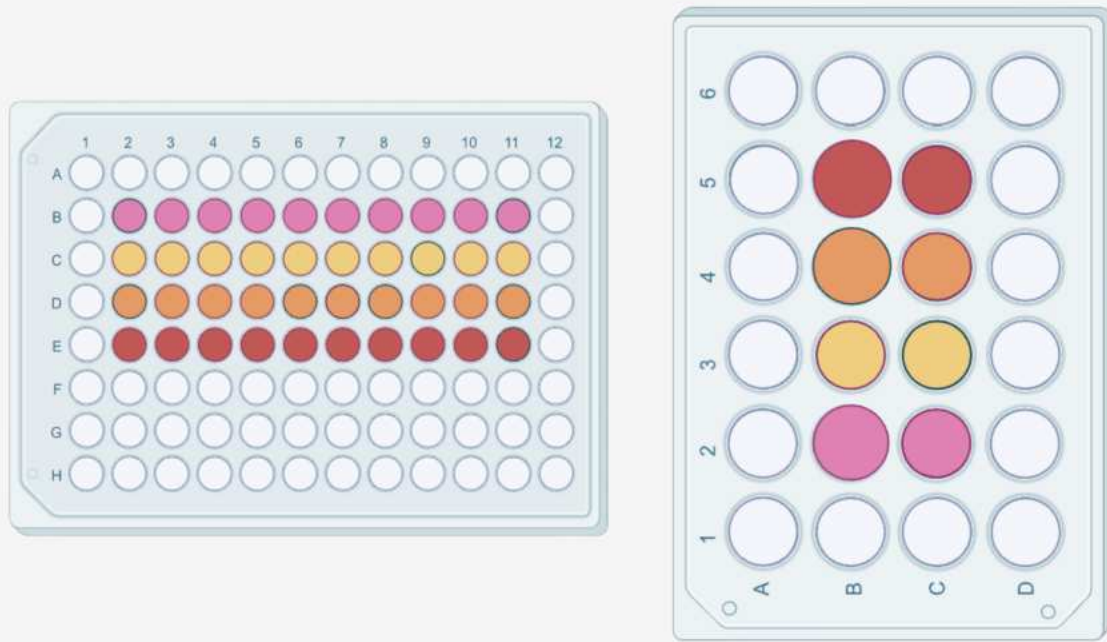


Figure 1. Raji Cells at low vs. high confluency
Images taken by researchers, 2026

2. Seed Raji Cells



96 well plate = Cell Titer Glo assay
24 well plate = Western Blot

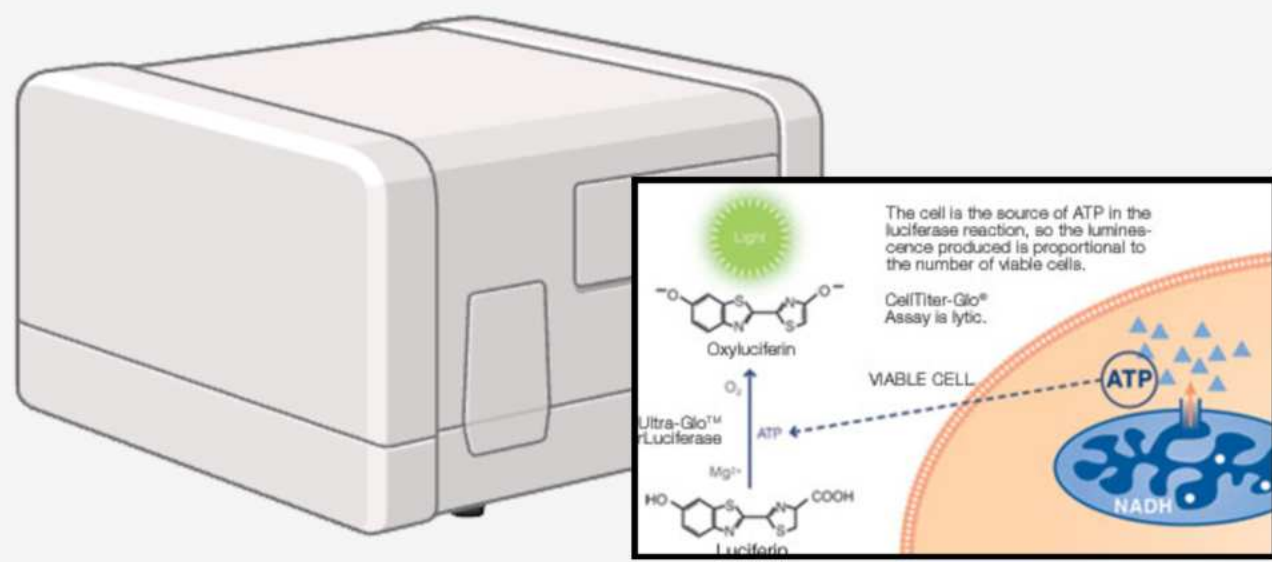
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3. Administer Treatments

- Negative Control:** 0µm ascorbyl palmitate + media
- Treatment 1:** 50µm ascorbyl palmitate + media
- Treatment 2:** 100µm ascorbyl palmitate + media
- Treatment 3:** 200µm ascorbyl palmitate + media

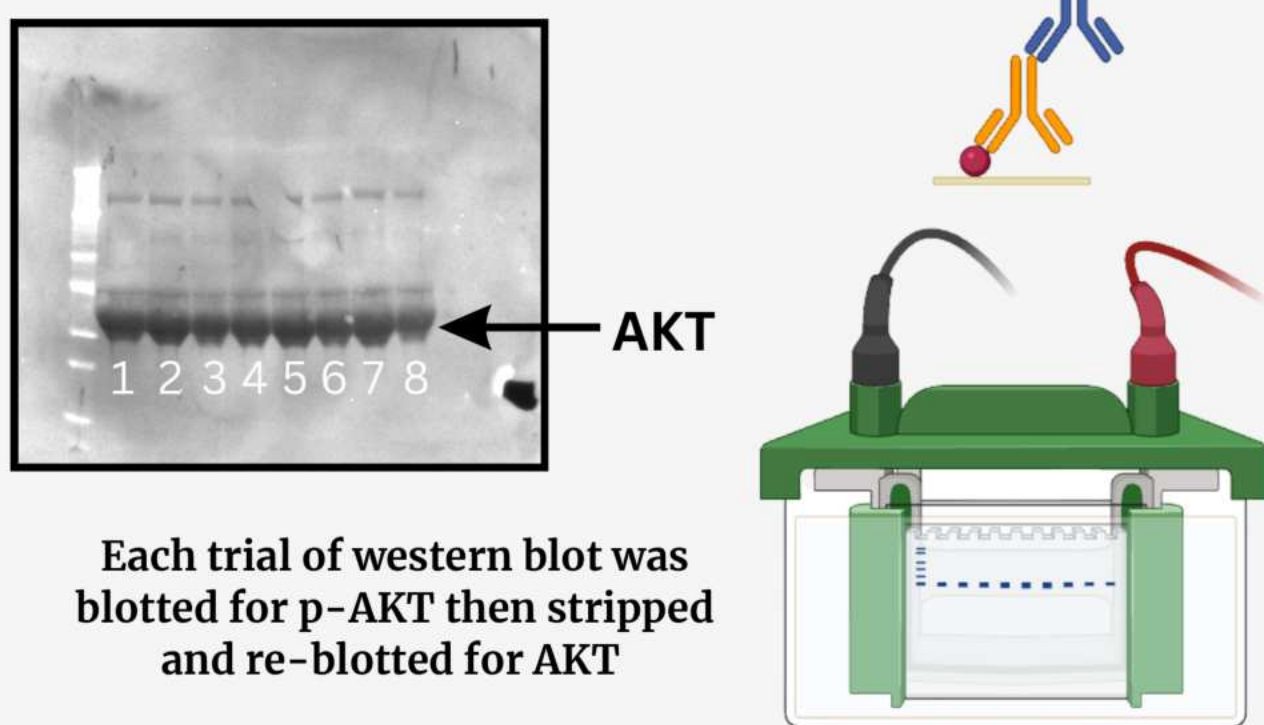
IV: Concentrations of Ascorbyl Palmitate
50µM, 100µM, 200µM
DV: Expression of Phosphorylated Akt Protein

4. Cell Titer Glo Assay



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https://www.promega.com/products/cell-health-assays/cell-viability-and-cytotoxicity-assays/celltiter_glo-luminescent-cell-viability-assay/

5. Western Blot

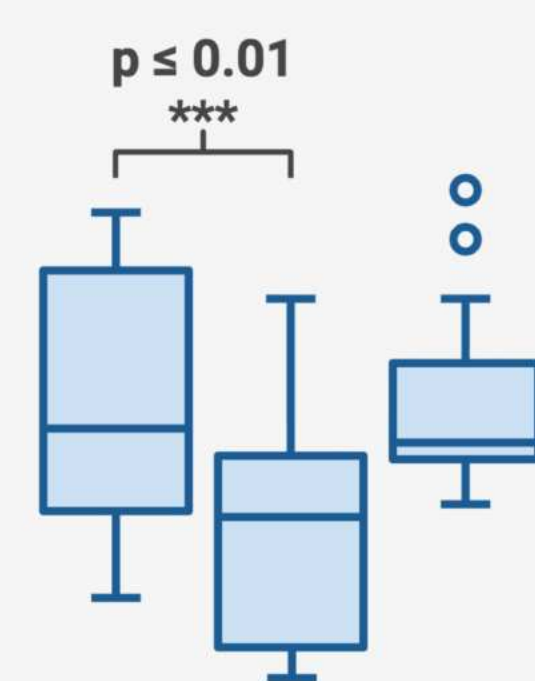


Each trial of western blot was blotted for p-AKT then stripped and re-blotted for AKT

Figure 2. Trial 2 Western Blot for AKT
(1. NC 0µm, 2. NC 0µm, 3. 50µm, 4. 50µm, 5. 100µm, 6. 100µm, 7. 200µm, 8. 200µm)

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6. Statistical Analysis



Parametric: Anova Statistical Test for both Cell titer & Western blot

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Results

Cell Titer Glo Assay

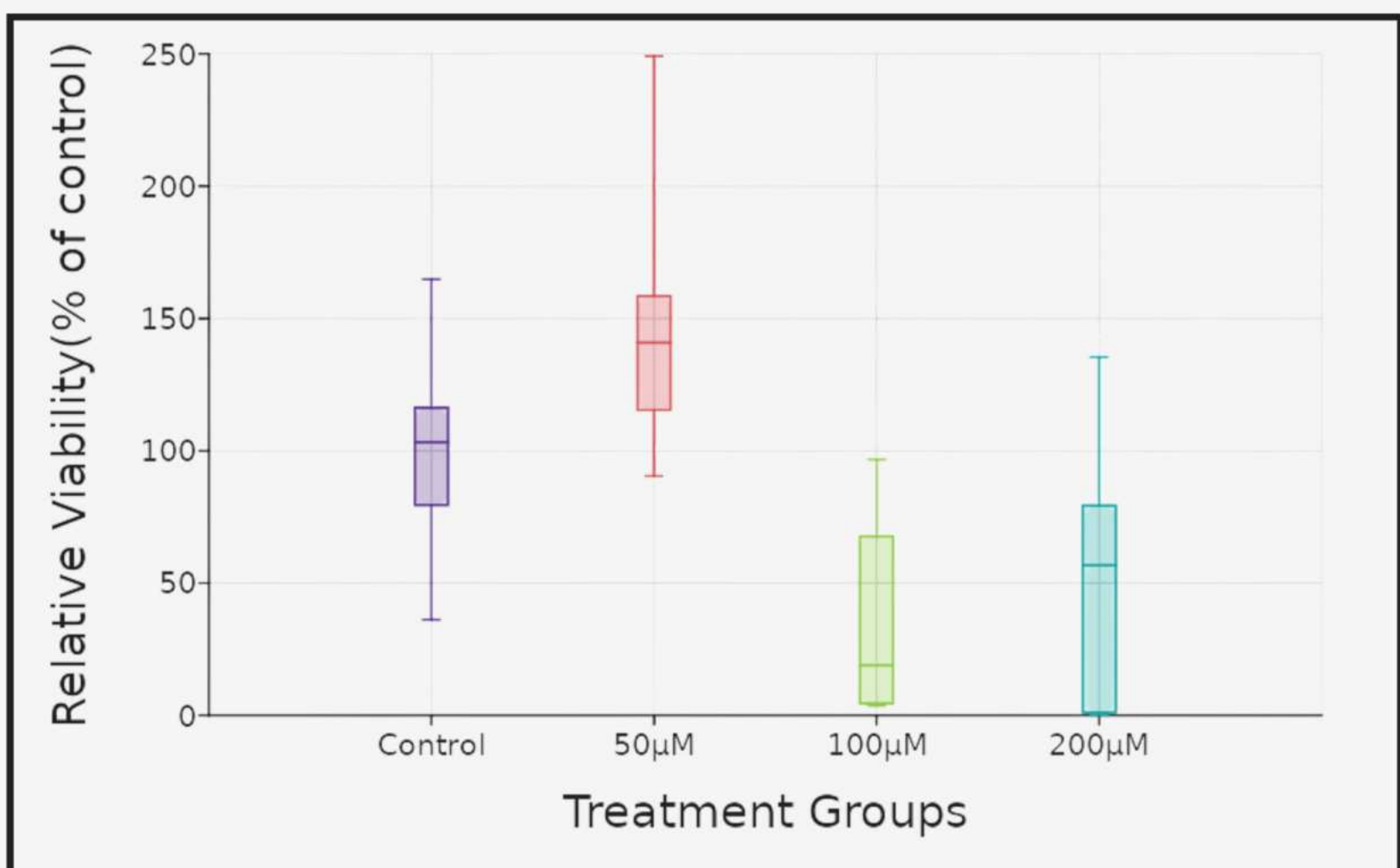


Figure 3. Median Cell viability(Relative Luminescence) Vs. Treatment groups

Data Visualization created by researchers using DataClassroom, 2026

ANOVA Results

Groups with Significant Differences	P-Values	ANOVA P-Value
50µM - Control	<0.01	< 0.01 (A P-value of <0.01 means extremely strong confidence that the groups are different.)
100µM - Control	<0.01	
200µM - Control	<0.01	
100µM - 50µM	<0.01	
200µM - 50µM	<0.01	
200µM - 100µM	<0.01	

Table 1. Significant ANOVA p-values comparing cell viability across different treatment and control groups.

Western Blot for p-AKT protien expression

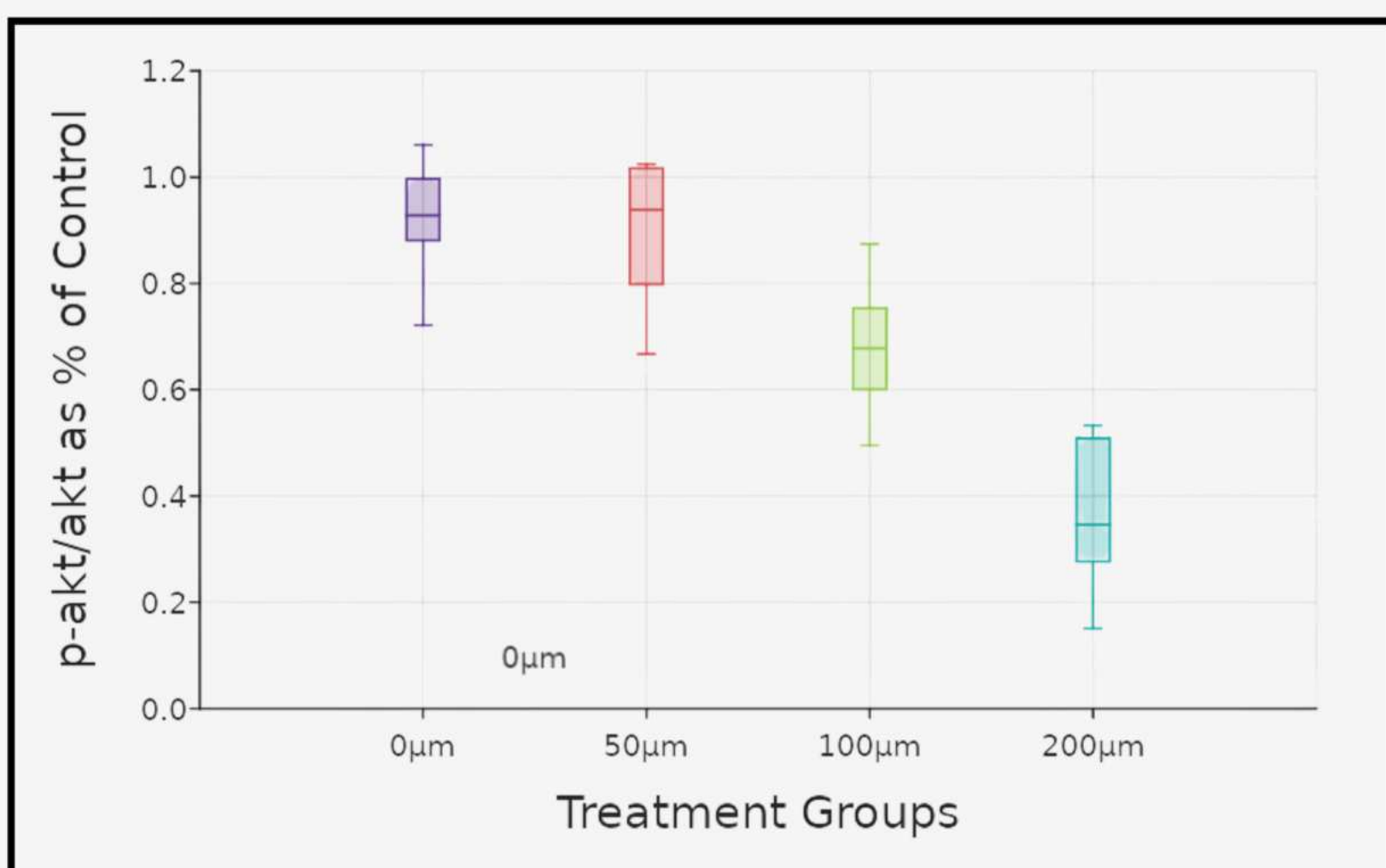


Figure 4. Median p-AKT/AKT as % of Control Vs. Treatment groups

Data Visualization created by researchers using DataClassroom, 2026

ANOVA Results

Groups with Significant Differences	P-Values	ANOVA P-Value
50µM - Control	0.99	< 0.01 (A P-value of <0.01 means extremely strong confidence that the groups are different.)
100µM - Control	0.03	
200µM - Control	<0.01	
100µM - 50µM	0.05	
200µM - 50µM	<0.01	
200µM - 100µM	<0.01	

Table 2. Significant ANOVA p-values comparing p-AKT expression across different treatment and control groups.

Discussion

Collected Data **Supported Hypothesis:**

- Higher concentrations of Ascorbyl palmitate led to a **decrease** in the **expression of p-AKT**.

Cell Titer Glo for Cell Viability:

- Cell Viability decreased significantly in a dose dependent pattern across all 3 trials
- 100µm and 200µm showed the greatest reduction in viability compared to the control
- 50µm showed minimal effect suggesting that the therapeutic affect of Ascorbyl Palmitate is above this concentration
- Conclusion: ANOVA p<0.01 across all trials conforms extremely strong statistical confidence that the treatment groups differ significantly from the control.**

Western Blot for p-AKT expression

- Normalized p-AKT/AKT declined over all 3 trials in a dose dependent pattern
- Greatest decrease observed at 200µm aproximatley ~60% reduction compared to the untreated control
- 50µm showed miniaml changes which is consitent with the cell viability results
- Conclusion: Confirm that Ascorbyl palmitate reduces AKT phosphorylation rather than degradeing the AKT protien.**

Implications

- Ascorbyl palmitate's ability to **supress** the PI3K-AKT pathway suggests that it could serve as a **less toxic alternative** to chemotherapy for Burkitts Lymphoma patients.
- Beacuse Ascorbyl palmitate is already **approved** as a food antioxidant it is already ahead in its potential **path to clinical usage**
- If effective in future in vivo models. Ascorbyl palmitate could provide a more **accessible** and **affordable** treatment for individuals with Burkitt's lymphoma.

Future Work

- Test Ascorbyl palmitate in **other lymphoma or cancer cell lines**
- Measure **alternative** pathways to **confirm** the mechanism
- Adjust** concentrations and/or treatment times based on results
- Test Ascorbyl palmitate in a **dual** treatment to see if it can **enhance** the effects of other treatments

References

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